

CareTrace

A provenance-first longitudinal care note that compresses fragmented interactions for action while keeping every AI-supported highlight anchored to immutable evidence.

React / Vite	REST / JSON + signed token	FastAPI + RBAC
Fast UI	Clinic scope on every call	Service + validation
Precomputed Glance	SQLAlchemy	Encrypted SQLite



Why this shape

Healthcare collaboration needs compression, but trust fails when a summary cannot show where it came from. CareTrace therefore treats AI output as a candidate, not a fact. The backend validates evidence, owns offsets, stores provenance, and may abstain. Page loads read a precomputed snapshot and never wait for an LLM.

Trust boundary

Raw interactions are committed and encrypted before AI work. Known names, phones and IC/ID patterns are redacted and checked before a provider call. Provider failures preserve the source and previous Glance. Logs and audit events contain metadata only.

Provider boundary

Ollama chat and embedding calls sit behind explicit provider interfaces. AI_PROVIDER=ollama|fixture is selected at startup and never changes silently. The fixture provider makes automated tests and offline evaluation reproducible.

Synthetic data only. CareTrace is priority support, not diagnosis or clinical decision support.

Evidence before emphasis

Backend-owned provenance

The LLM returns `source_ref`, a verbatim `evidence_quote` and a normalized candidate - never character offsets. The backend searches only the referenced immutable source version. A unique exact match is verified; a unique Unicode/whitespace-normalized match is supported. Missing or repeated matches are `review_required` and cannot become automatic highlights.

Redaction creates a boundary map from redacted positions to the encrypted original. The stored `ProvenanceEdge` contains source entry/version, section, both coordinate systems, match method and a quote hash. Every read revalidates the hash. Later edits create new versions, so citations do not drift.

Relational spine

AUTHORITY	RECORD	TRUST + ACTION
Clinic User Patient	Interaction TimelineEntry EntrySection EntryVersion	ClinicalFact ProvenanceEdge Highlight GlanceSnapshot
Signed token Clinic scope	Role-owned sections Immutable snapshots	Comment / Conflict Feedback / Audit

Deterministic collaboration

Staff and clinicians own separate sections. Independent section counters allow simultaneous edits without overwriting each other; a stale same-section write returns 409. Revert restores one authorized section into a new version. Conflicting allergy or dosage facts enter a review queue; only a clinician can select the authoritative fact, and both sources remain.

Bounded adaptive ranking

final = max(applicable risk floor, clamp(rule score + learned bonus, 0, 100))

Risk floors are intentionally narrow: explicit synthetic risk tag, allergy conflict, dosage conflict, critical unresolved action or clinician-confirmed warning. Generic symptoms never create a floor. Clinical entities affect ordering only, not diagnosis.

Explicit positive and negative feedback updates per-user embedding centroids. The similarity bonus is clamped to +/-8; non-interaction is ignored; critical floors cannot be suppressed.

Know when the system is wrong

VERIFIED	SUPPORTED	ABSTAIN
Unique exact quote Schema valid	Unique normalized match Visible label	Missing / repeated quote No automatic highlight

Security and patient boundary

Passwords use scrypt and tokens use HMAC-SHA256. Every query rechecks subject, role and clinic against SQLite. Patients are linked to one patient ID and can read only clinician-approved summary/instruction rows; timeline, Glance, provenance, versions, comments and audit endpoints reject their token. Local HTTP is for demonstration only; deployment requires TLS termination and managed secrets.

Automated evaluation

The required five micro-test modules cover cross-role writes, patient exclusions, clinic isolation, version increments, revert, metadata-only audit, immutable provenance, concurrent sections, deterministic 409 conflicts and adaptive importance. Additional tests verify that LLM offsets are ignored, ambiguous quotes abstain, PHI never reaches a spy provider, ordinary symptom text creates no risk floor and critical items survive rejection.

1.16 ms	1.44 ms	< 300 ms
median	P95	target

Warm-path approximation: local in-process HTTP client, 500 timeline entries, 20 warm-up requests and 500 measured GET /glance requests. This isolates application/database time and excludes proxy, network and production concurrency effects.

Scope and migration

Configurable prototype defaults are DATA_DECAY_THRESHOLD_DAYS=90, RECENT_CONTEXT_COUNT=3, RAG_TOP_K=5 and GLANCE_MAX_ITEMS=5. Voice, cloud deployment, real EHR integration and formal compliance are out of scope. A production migration would add a durable queue, managed database, key rotation, operational telemetry, extraction datasets and reviewed clinical governance before real patient use.

DESIGN PRINCIPLE	Compression may be probabilistic. Provenance, permissions and overrides are deterministic.
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